

Airline Passenger Satisfaction Prediction Using Data Mining Techniques

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STRUCTURED ABSTRACT

Objective: This study investigates the prediction of airline passenger satisfaction using supervised data mining techniques. The aim is to identify key factors influencing satisfaction and compare the predictive performance of Decision Tree and Random Forest models.

Dataset: The analysis uses a dataset of 5,198 airline passenger records containing demographic information, travel details, service ratings and delay information. The target variable is satisfaction, a binary class indicating whether a passenger was satisfied or neutral/dissatisfied.

Methods: Data preprocessing included removing invalid observations, handling missing values and encoding categorical features using One-to-Many encoding. Two supervised learning techniques were applied: a Decision Tree and a Random Forest. Both models were evaluated using stratified 5-fold cross-validation. Performance metrics included accuracy, precision, recall, F1 score and area under the receiver operating characteristic curve (AUC).

Results: The Random Forest achieved higher accuracy and AUC than the Decision Tree, while the Decision Tree provided interpretable insights into influential predictors. Passenger class and service quality variables were the most influential factors in predicting satisfaction.

Conclusions: Ensemble methods improve predictive performance, whereas interpretable models offer actionable insights for airline service improvement. The study demonstrates the importance of travel class and service ratings in shaping passenger satisfaction.

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1 INTRODUCTION

1.1 CONTEXT AND DATASET

The dataset used in this study is the Airline Passenger Satisfaction Dataset, derived from a modified Kaggle dataset. It contains 5,198 records representing individual passengers, with 22 attributes capturing demographic information, travel patterns and service ratings. The target variable, satisfaction, is binary, indicating whether a passenger was satisfied or neutral/dissatisfied with their flight experience. The dataset shows a slight class imbalance, with 2,022 passengers reporting satisfaction and 2,594 reporting neutral or dissatisfied experiences. Key features include passenger class, travel type, loyalty status and ratings of seat comfort, inflight entertainment, food and drink, on-board service and cleanliness. Additional variables capture flight delays, online check-in usage and baggage handling service, providing a comprehensive view of passenger experiences.

1.2 PREDICTION TASK

The objective of this analysis is to predict passenger satisfaction based on available features. This is formulated as a supervised classification problem, where input variables estimate the likelihood of a passenger being satisfied. Accurate prediction supports airlines in identifying service gaps and improving passenger retention. Preprocessing steps, including handling imbalanced classes, cleaning missing or anomalous data and encoding categorical variables make the dataset is suitable for reliable predictive modelling (Werner de Vargas et al., 2022).

1.3 OVERVIEW OF METHODS

Two supervised learning methods were applied: Decision Tree and Random Forest. Decision Trees provide interpretable models by recursively partitioning the dataset based on feature values, revealing the most influential predictors (Song & Lu, 2015). Random Forests, ensembles of Decision Trees, improve stability and predictive performance by aggregating multiple trees and reducing variance (Salman et al., 2024). Both models were implemented in KNIME 5.3.3 and evaluated using stratified 5-fold cross-validation to make the evaluation more robust (Wilimitis & Walsh, 2023). Performance metrics included accuracy, precision, recall, F1 score and the area under the receiver operating characteristic curve (AUC).

2 DATA MINING THEORY

2.1 DECISION TREE

A Decision Tree is a supervised machine learning algorithm for classification and regression tasks that builds a model through recursively partitioning the dataset into subsets based on feature values. Each internal node represents a decision rule on a single attribute, each branch represents the outcome of that rule and each leaf node represents the predicted class. The algorithm selects the attribute that best separates the classes at each step, typically using impurity measures such as Gini impurity or information gain derived from entropy. A split reduces impurity when resulting subsets contain observations predominantly from a single class. The process continues until a stopping condition, such as a maximum tree depth or minimum records per node, is reached, preventing overfitting (Song & Lu, 2015).

Decision Trees are widely used because they provide interpretable models (Song & Lu, 2015). The tree structure highlights which attributes contribute most to classification. The attribute at the root node is usually the most influential, producing the largest class separation (Hajjej et al., 2022). In customer satisfaction analysis, Decision Trees offer a transparent way to identify influential features that drive predictions. In this study, a Decision Tree provides a baseline model for predicting airline passenger satisfaction, revealing the most important service attributes.

2.2 RANDOM FOREST

Random Forest is an ensemble learning method that extends Decision Trees. Rather than constructing a single tree, the algorithm builds multiple Decision Trees and aggregates their predictions to produce a final classification. Each tree is trained on a random subset of the data and a random subset of the features, introducing diversity and reducing correlation among trees (Salman et al., 2024).

Ensembling improves stability and predictive performance (Salman et al., 2024). Individual Decision Trees have high variance, meaning small changes in the training data can substantially alter the model (Kern et al., 2019). Random Forest reduces this variance by averaging predictions across many trees, producing lower variance and better generalisation (Salman et al., 2024).

Random Forest can model complex, non-linear relationships without strong assumptions about data distribution (Auret & Aldrich, 2012). This makes it suitable for datasets with many service ratings and behavioural variables, such as airline passenger satisfaction data. Its ability to handle numerical and categorical inputs, combined with strong predictive performance, has made Random Forest a widely adopted benchmark algorithm (Breiman, 2001).

2.3 Evaluation Metrics

Assessing predictive models requires quantitative metrics that measure classification performance. For binary classification problems, common metrics include accuracy, precision, recall, F1 score and AUC (Rainio et al., 2024).

Accuracy measures the proportion of correctly classified observations relative to the total predictions (Hicks et al., 2022). While intuitive, it can be misleading with imbalanced class distributions (Aguilar-Ruiz & Michalak, 2024). Precision measures the proportion of predicted positive observations that are correct, while recall measures the proportion of actual positives correctly identified (Hicks et al., 2022). The F1 score combines precision and recall into a single metric via their harmonic mean, balancing the impact of false positives and false negatives (Hicks et al., 2022).

The receiver operating characteristic (ROC) curve plots the true positive rate against the false positive rate across different classification thresholds (Nahm, 2022). The AUC summarises the model's ability to discriminate between classes, with higher values indicating stronger discriminatory power (Nahm, 2022).

To obtain reliable performance estimates, models in this study were evaluated using stratified cross-validation. This divides the dataset into multiple folds while preserving class distributions. Each fold serves as a test set once, while the remaining folds train the model. Cross-validation reduces overfitting risk and provides a more robust estimate of performance on unseen data (Wilimitis & Walsh, 2023).

3 Data Exploration and Preparation

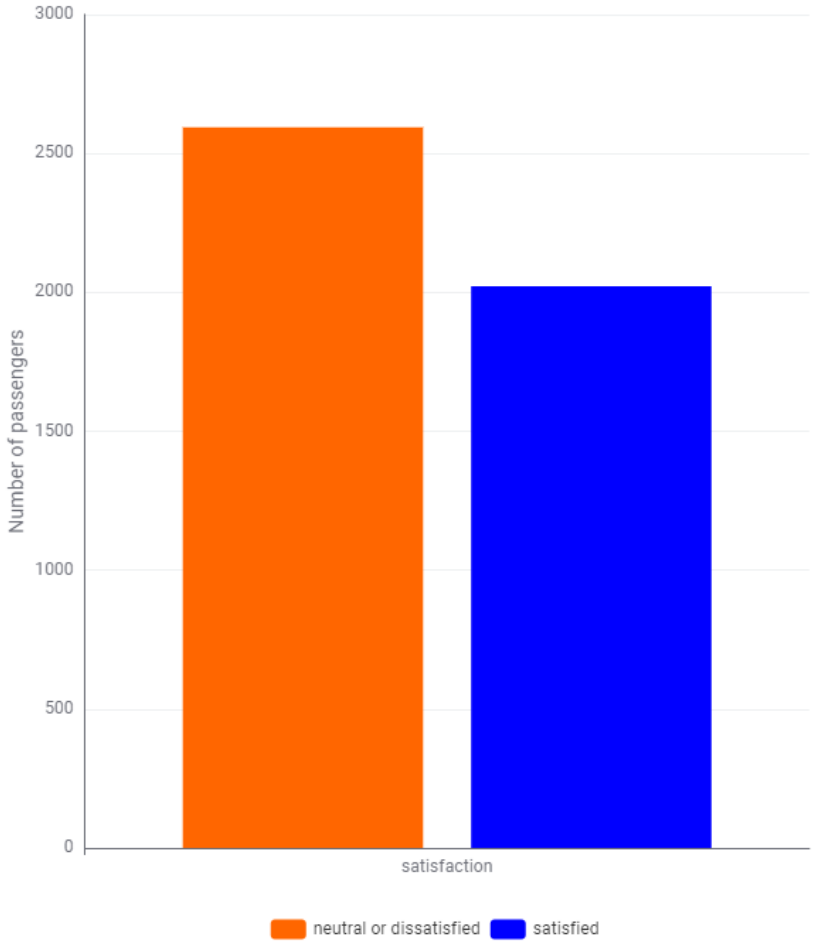
3.1 Initial Data Inspection

The dataset was imported into KNIME and inspected to assess structure and variable distributions. It contained 5,198 passenger records and 22 attributes describing demographics, travel behaviour, service ratings and flight delay information. The target variable, satisfaction, is a binary class indicating whether a passenger reported being satisfied or neutral/dissatisfied.

Class distribution revealed a small imbalance, with 2,022 satisfied and 2,594 neutral/dissatisfied passengers (Figure 1). This imbalance necessitates careful evaluation to avoid biased performance metrics.

Figure 1

Distribution of passenger satisfaction classes in the dataset



Note. The blue bar represents passengers who reported being satisfied and the orange bar represents passengers who were neutral or dissatisfied.

Service ratings were ordinal, ranging from 0 to 5, capturing seat comfort, inflight entertainment, food and drink, cleanliness and on-board service. Delay variables measured departure and arrival delays in minutes. Demographic features included age, gender, loyalty status, travel purpose and travel class. Initial inspection identified anomalies, such as age values of 999 and missing entries across several attributes, requiring correction during preprocessing.

3.2 Data Cleaning

Data cleaning prepared the dataset for predictive modelling. The identifier column was removed as it did not contribute predictive value. Invalid records, such as unrealistic age values above 120, were filtered out to maintain plausibility.

Missing values were addressed using KNIME's Missing Value node. Numeric attributes were imputed with the column mean, while categorical attributes were imputed with the most frequent category. This approach maintained dataset size while reducing the impact of missing data on model performance. These steps produced a dataset with valid, complete records ready for feature transformation and model training.

3.3 Feature Encoding

Several variables were categorical, including gender, loyalty status, travel type and travel class. Many machine learning algorithms require numeric inputs, necessitating conversion of categorical attributes.

One-to-Many encoding was applied in KNIME, transforming each category into a separate binary column representing its presence or absence. For example, travel class generated binary columns for Business, Eco and Eco Plus classes. Other categorical variables, such as gender and travel type, were similarly encoded. This method preserved categorical information while producing a fully numeric dataset suitable for machine learning.

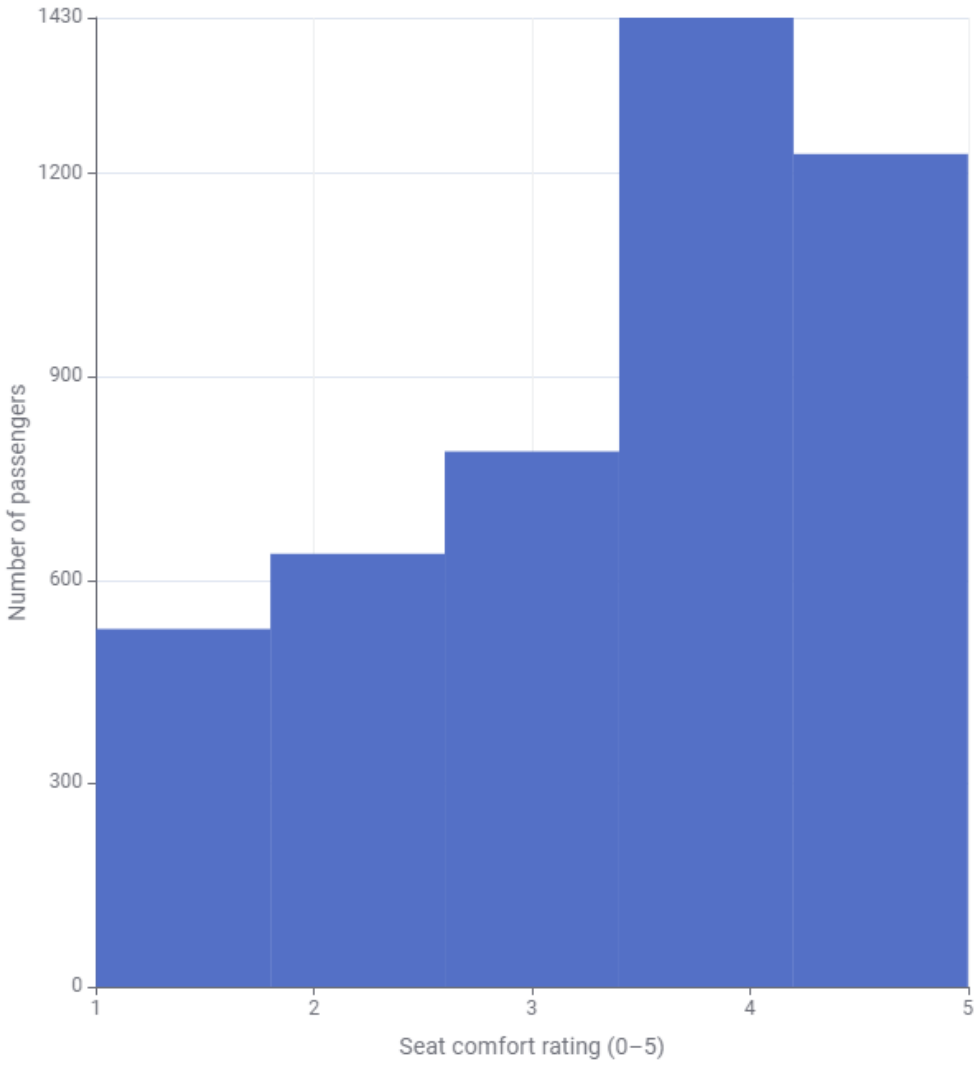
3.4 Exploratory Visuals

Exploratory data analysis assessed patterns and relationships between features and satisfaction. A bar chart confirmed the slight class imbalance observed in the initial inspection. Histograms of service ratings, including seat comfort (Figure 2), inflight entertainment (Figure 3) and on-board service (Figure 4), showed concentration around higher ratings, with noticeable variation between passengers. A scatter plot of departure versus arrival delays (Figure 5)

indicated a strong positive relationship, though service ratings and travel class showed stronger associations with satisfaction.

Figure 2

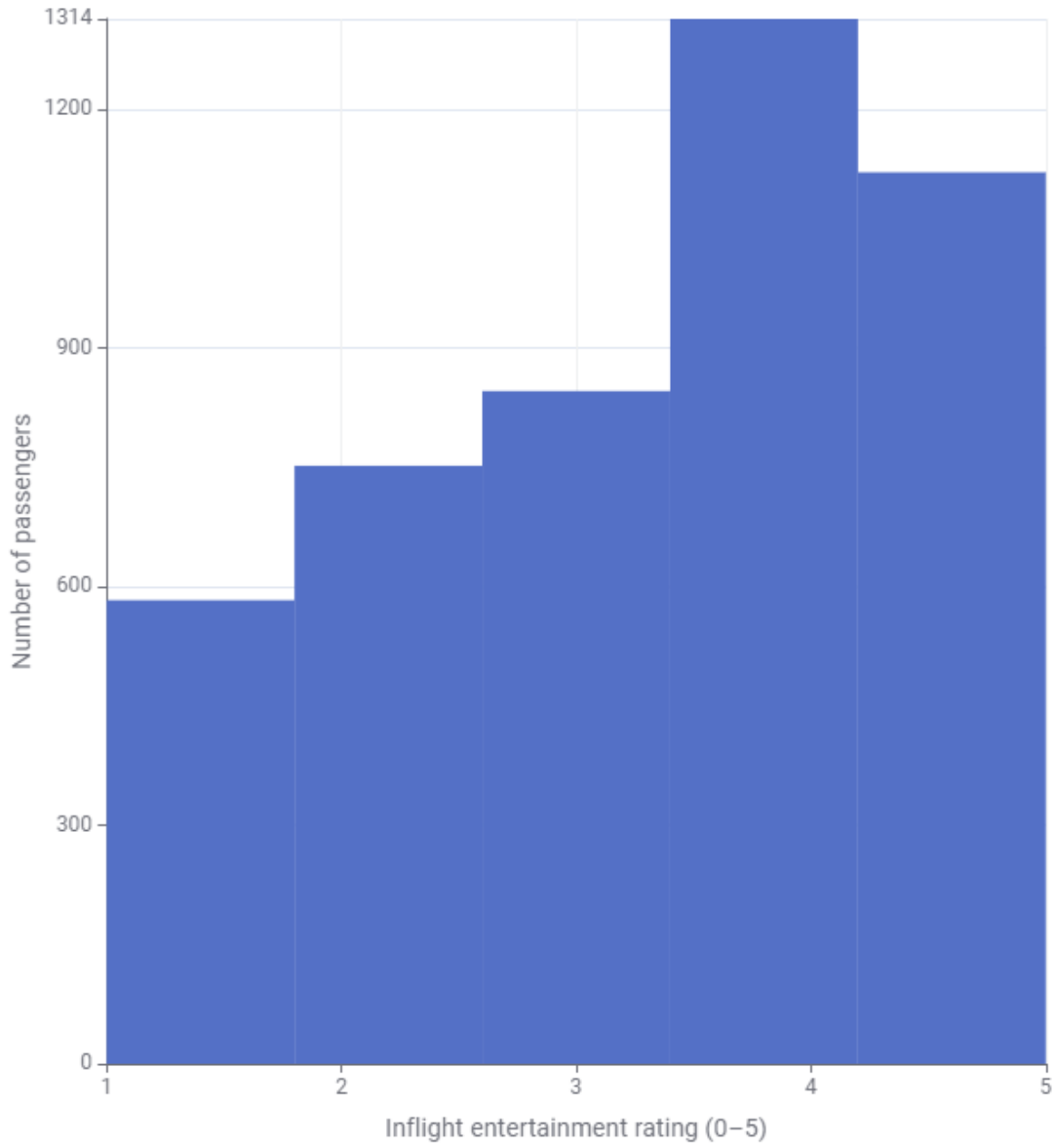
Distribution of seat comfort ratings among airline passengers



Note. Ratings range from 0 (lowest) to 5 (highest), showing how the passengers rated the seating comfort.

Figure 3

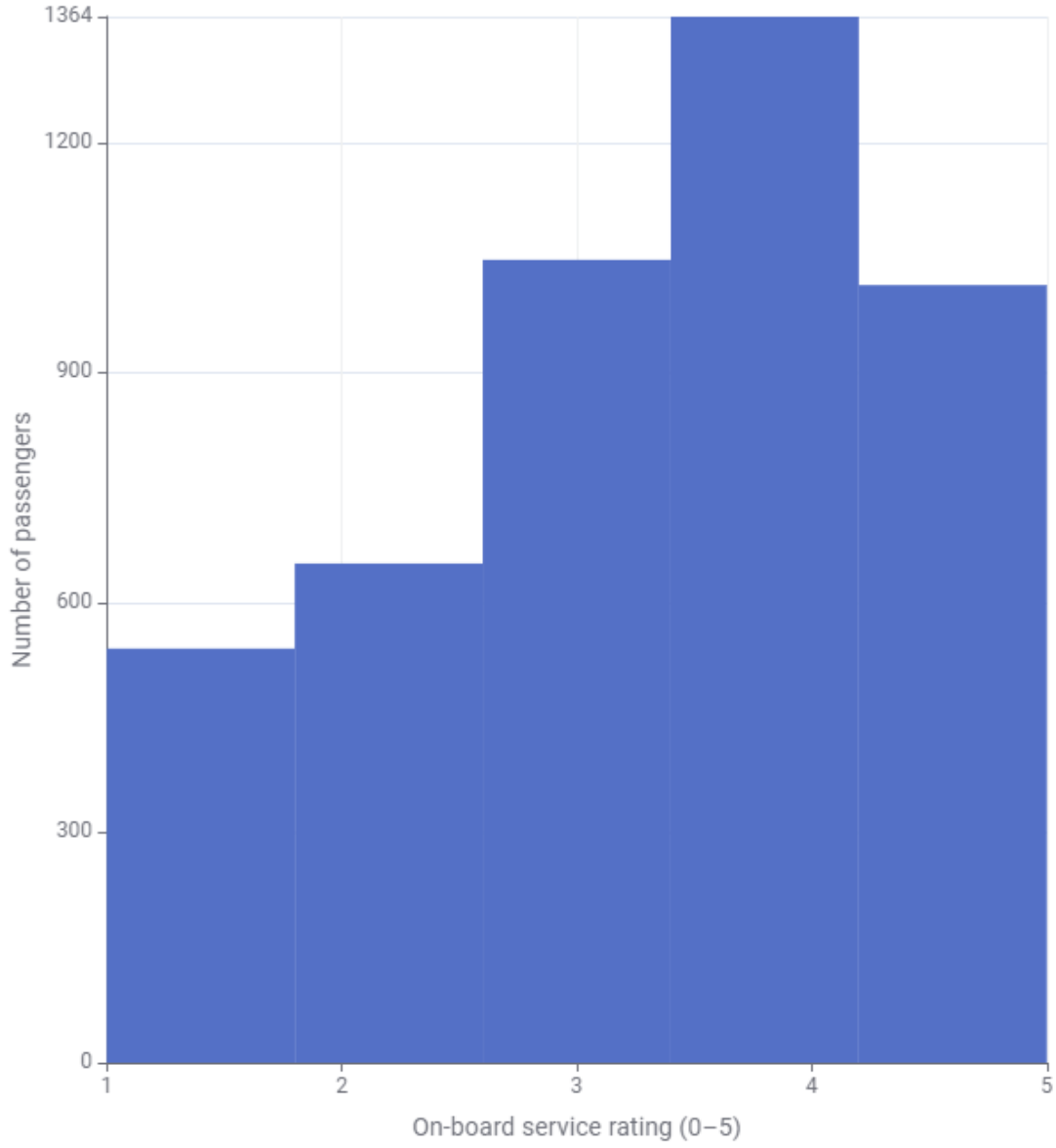
Distribution of inflight entertainment ratings among airline passengers



Note. Ratings range from 0 (lowest) to 5 (highest), showing how the passengers rated the entertainment service.

Figure 4

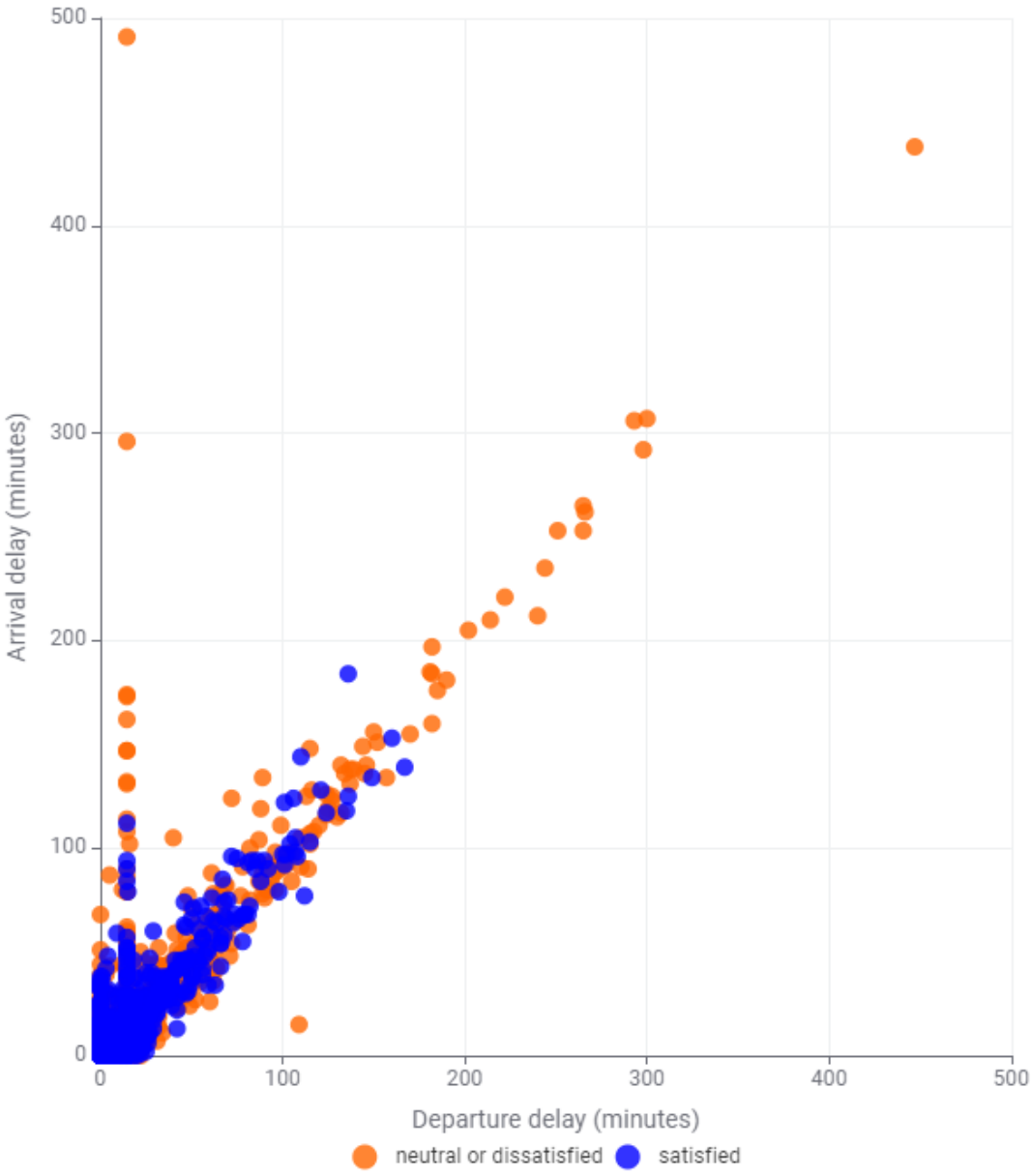
Distribution of on-board service ratings



Note. Ratings range from 0 (lowest) to 5 (highest), showing how the passengers rated the on-board service.

Figure 5

Relationship between departure delay and arrival delay

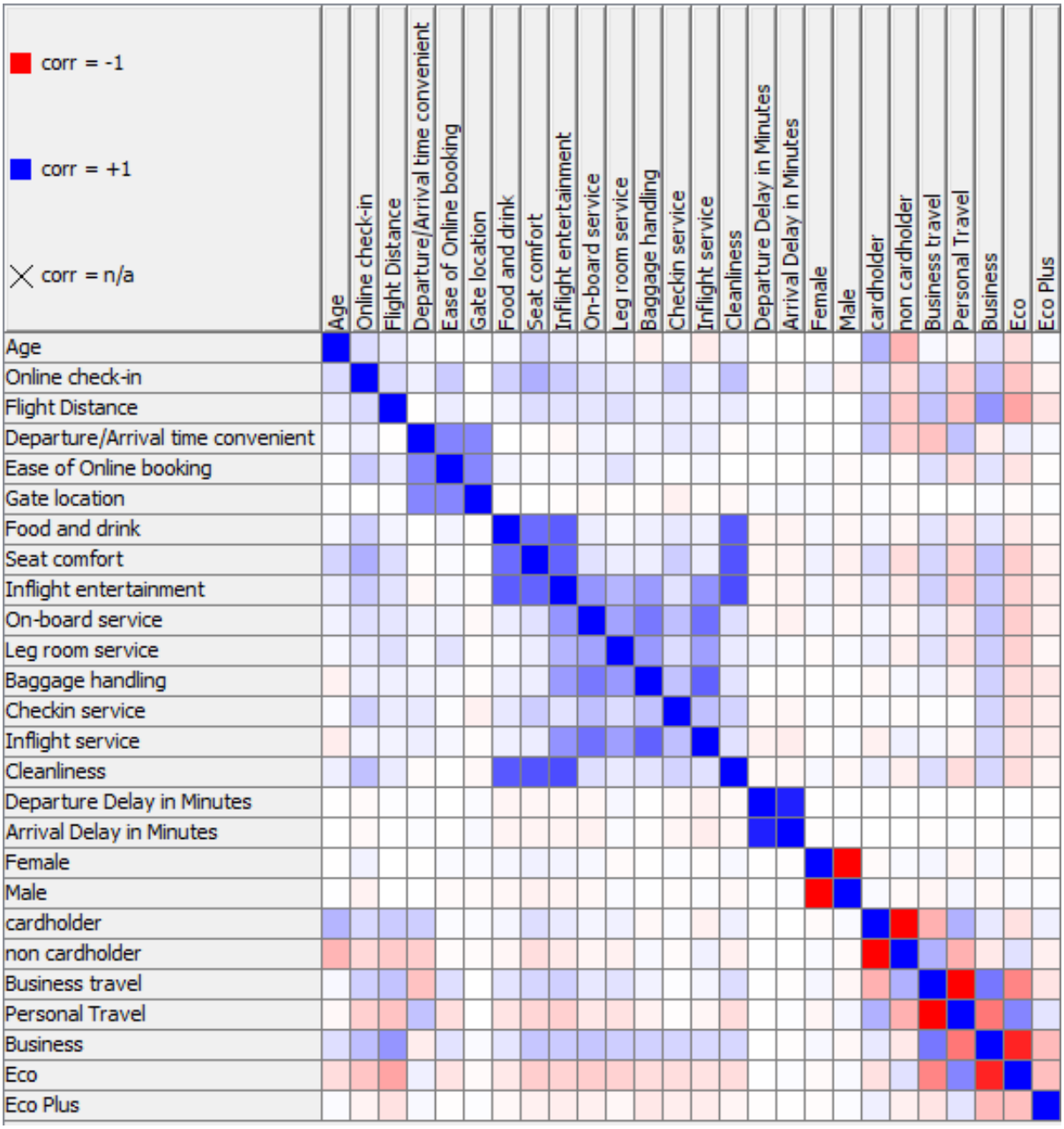


Note. Points are coloured by satisfaction class: blue for satisfied passengers, orange for neutral or dissatisfied passengers.

Correlation analysis among numeric attributes revealed moderate positive correlations between service quality variables (Figure 6). Passengers rating one service aspect highly often rated others highly, suggesting potential multicollinearity that may influence model behaviour.

Figure 6

Correlation heatmap of key service rating variables



Note. Blue indicates positive correlation, red indicates negative correlation and white represents no correlation.

These analyses provided insight into dataset structure, guided feature selection and informed preprocessing decisions for the modelling stage.

4 EXPERIMENTAL SETUP

4.1 CROSS-VALIDATION

A 5-fold stratified cross-validation was implemented in KNIME using the X-Partitioner node to provide robust evaluation (Wilimitis & Walsh, 2023). Stratification maintained the original proportion of satisfied and neutral/dissatisfied passengers in each fold, addressing class imbalance (Kernan et al., 1999). Each fold was used once as a test set, with the remaining four folds serving as training data. Predictions from all folds were aggregated using the X-Aggregator node, producing a single set of predictions for evaluation. This approach reduces overfitting and provides a reliable estimate of model performance on unseen data, consistent with best practices in predictive modelling (Wilimitis & Walsh, 2023).

4.2 DECISION TREE CONFIGURATION

The Decision Tree model was developed using the Decision Tree Learner node. The target variable was satisfaction and all preprocessed numeric features from One-to-Many encoding were included. Key parameters were a maximum tree depth of 10 and a minimum of 5 records per node. These settings constrained tree complexity while allowing sufficient granularity to capture patterns in passenger satisfaction (Azad & Moshkov, 2023).

Predictions were generated with the Decision Tree Predictor node. Inspection of the resulting tree revealed that the root node split on passenger class, confirming its dominant predictive influence. Secondary splits involved service quality variables such as seat comfort and inflight entertainment, illustrating the model's ability to highlight important attributes. This transparency supports interpretability, an advantage of Decision Trees for operational decision-making in airline services.

4.3 RANDOM FOREST CONFIGURATION

Random Forest modelling was implemented using the Random Forest Learner node, with 100 trees and a maximum depth of 15 for individual trees. At each split, a random subset of features was selected to reduce correlation between trees and improve generalisation. All preprocessed numeric features were included as input variables.

Predictions were generated with the Random Forest Predictor node and combined using the X-Aggregator node. The ensemble approach reduces variance and stabilises predictions compared with a single Decision Tree, particularly for observations with less common feature combinations. This aligns with literature showing that Random Forests improve accuracy and robustness through aggregation of multiple trees (Breiman, 2001).

4.4 Evaluation Nodes in KNIME

Model performance was assessed using the Scorer and ROC Curve nodes. The Scorer node produced confusion matrices and standard classification metrics, while the ROC Curve node generated ROC curves and calculated AUC. These nodes enabled a clear, reproducible comparison of model performance without presenting numeric results in this section.

The workflow combines preprocessing, One-to-Many encoding, stratified cross-validation and carefully configured learner and predictor nodes. This design supports reproducibility, interpretability and robust evaluation, meeting the requirements of rigorous predictive modelling in data mining tasks.

5 RESULTS AND DISCUSSION

5.1 PERFORMANCE COMPARISON TABLE

Model performance was measured using accuracy, precision, recall, F1 score and AUC. Table 1 summarises the results.

Table 1

Performance metrics for Decision Tree and Random Forest

Model	Accuracy	Recall (N/D)	Recall (S)	Precision (N/D)	Precision (S)	F1 Score	AUC
Decision Tree	0.888	0.911	0.858	0.892	0.883	0.870	0.935
Random Forest	0.902	0.948	0.843	0.886	0.927	0.899	0.969

Note. S = satisfied passengers, N/D = neutral or dissatisfied passengers.

The Random Forest shows improved overall accuracy and AUC relative to the Decision Tree, reflecting reduced variance through ensemble averaging.

5.2 CONFUSION MATRICES

Confusion matrices provide detailed insight into classification outcomes. The Decision Tree and Random Forest confusion matrices are shown in Tables 2 and 3.

Table 2

Confusion matrix for the Decision Tree model

	Predicted (N/D)	Predicted (S)
Actual (N/D)	2362	232
Actual (S)	287	1735

Note. S = satisfied passengers, N/D = neutral or dissatisfied passengers.

Table 3

Confusion matrix for the Random Forest model

	Predicted (N/D)	**Predicted (S)**
Actual (N/D)	2459	135
Actual (S)	317	1705

Note. S = satisfied passengers, N/D = neutral or dissatisfied passengers.

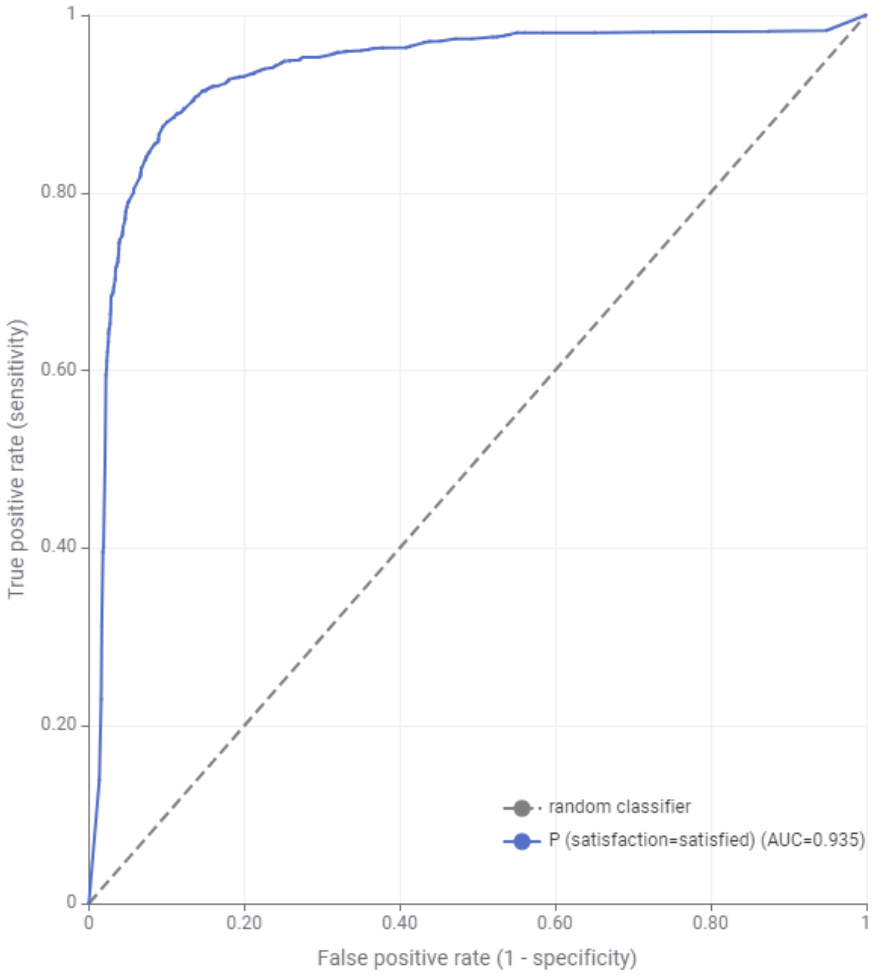
Random Forest reduced false negatives for neutral/dissatisfied passengers and false positives for satisfied passengers, aligning with the higher overall metrics observed.

5.3 ROC Curves

ROC curves visualised the models' discrimination ability. The Decision Tree achieved an AUC of 0.935 (Figure 7), indicating strong separation between satisfied and neutral/dissatisfied classes. Random Forest improved AUC to 0.969 (Figure 8), reflecting enhanced discriminatory power due to aggregation of multiple trees, which stabilises predictions and mitigates variance.

Figure 7

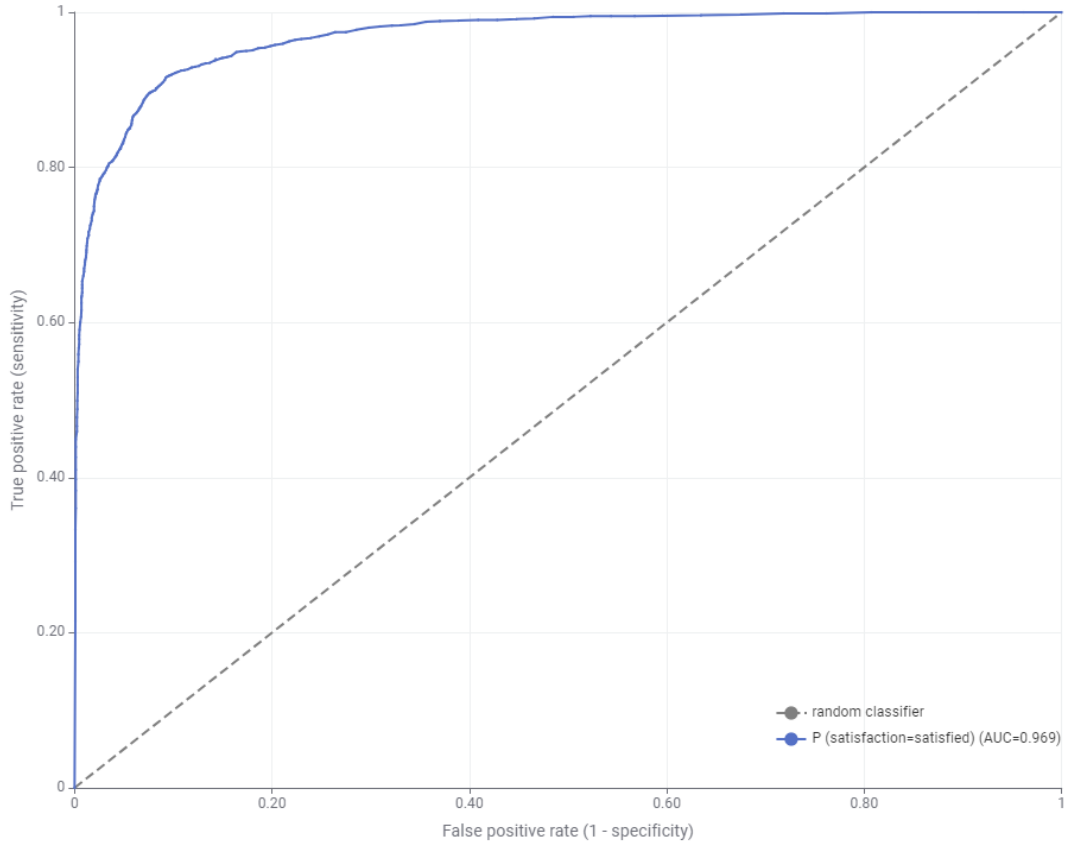
ROC curve for Decision Tree



Note. The blue curve represents the model's AUC, while the dotted diagonal line indicates a random classifier. The y-axis shows true positive rate (sensitivity) and the x-axis shows false positive rate.

Figure 8

ROC curve for Random Forest



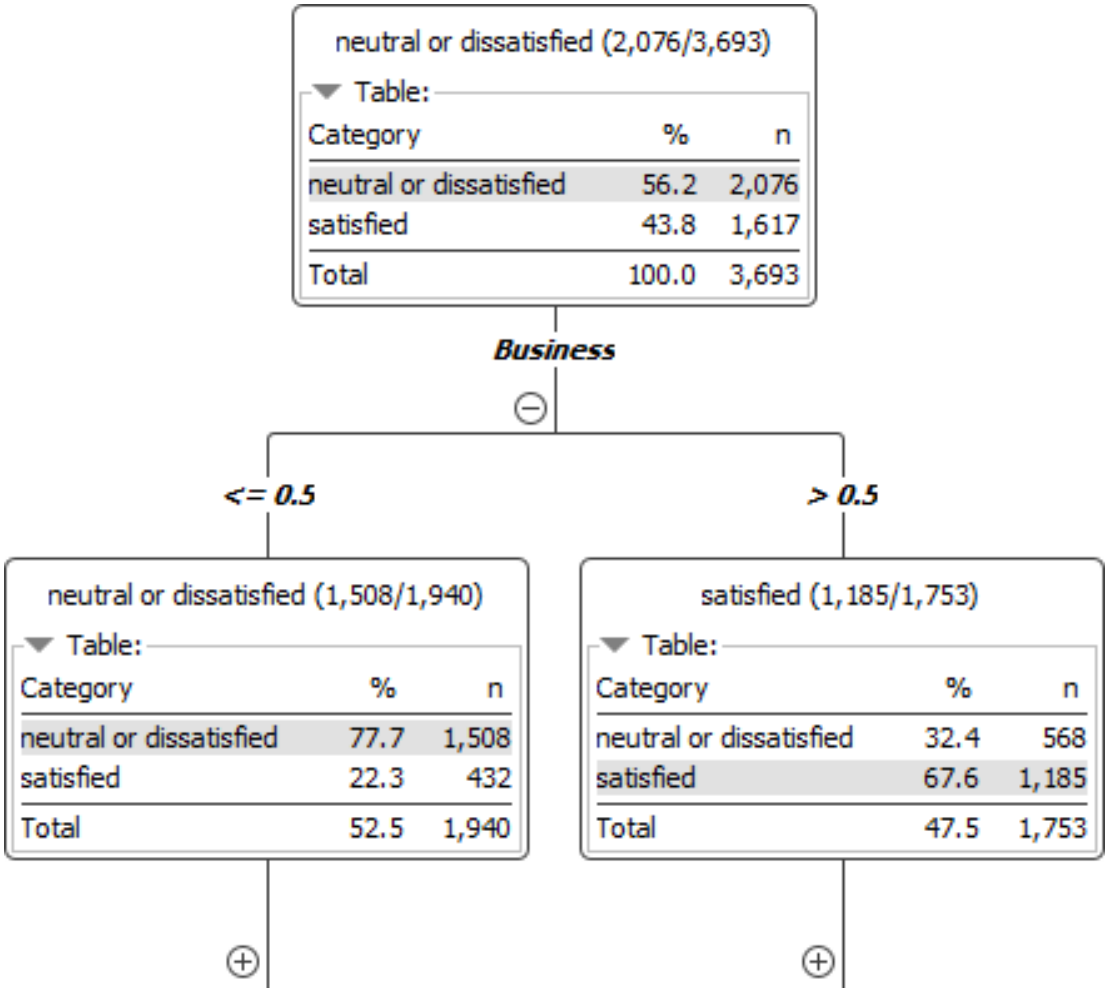
Note. The blue curve represents the model's AUC and the dotted diagonal line indicates a random classifier. The y-axis shows true positive rate (sensitivity) and the x-axis shows false positive rate.

5.4 Feature Importance

Feature importance was analysed from both model outputs. The Decision Tree identified passenger class as the root split, indicating that Business class passengers were more likely to report satisfaction (see Figure 9). Secondary splits included seat comfort, inflight entertainment, on-board service and cleanliness.

Figure 9

Decision Tree structure for passenger satisfaction classification



Note. The root node shows the most important split, with the next two levels of nodes representing secondary splits that further partition the data based on feature values.

Random Forest feature importance, derived from mean decrease in impurity, confirmed passenger class as the dominant predictor. Service-related variables remained consistently influential. Flight delays contributed minimally to model predictions compared with service quality features.

5.5 Discussion

Random Forest slightly outperforms the Decision Tree in accuracy and AUC, demonstrating the benefit of ensemble learning for this dataset, consistent with prior research (Breiman, 2001). However, the Decision Tree offers interpretable insights, identifying key factors affecting passenger satisfaction. Both models highlight travel class as the primary determinant, with service quality ratings playing a secondary but consistent role. The dataset size and source may limit generalisation to real-world airline environments.

Discrepancies in false positive and false negative predictions suggest that passenger satisfaction is affected by factors not captured in the dataset. The Random Forest achieved higher precision for satisfied passengers but slightly lower recall, indicating a more conservative classification for this class.

The results indicate that service quality and travel class are critical determinants of satisfaction. Future work could include hyperparameter tuning of the Random Forest, integration of additional demographic variables or evaluation against alternative classification algorithms. Interpretable models like Decision Trees remain valuable for actionable insights in operational decision-making, whereas ensembles maximise predictive performance.

6 CONCLUSION AND REFLECTIONS

6.1 SUMMARY OF FINDINGS

This study applied supervised data mining techniques to predict airline passenger satisfaction using a dataset of 5,198 records. Both Decision Tree and Random Forest models were implemented and evaluated using stratified 5-fold cross-validation.

Analysis revealed that passenger class was the strongest predictor of satisfaction, followed by service quality variables including seat comfort, inflight entertainment, on-board service and cleanliness. Flight delays contributed minimally to predictive performance.

Comparison of models indicated that Random Forest slightly outperformed the Decision Tree in accuracy and AUC, confirming the benefit of ensemble learning. The Decision Tree, however, offered greater interpretability, allowing identification of key features affecting satisfaction. ROC curves highlighted the superior discriminatory power of Random Forest, while confusion matrices showed that both models maintained reasonable performance across satisfied and neutral/dissatisfied classes.

6.2 METHOD REFLECTIONS

The Decision Tree provided a transparent baseline model. Its structure allowed clear identification of the most influential predictors, offering actionable insights for airline service improvements. The main limitation is sensitivity to variations in training data, which may result in overfitting to minor patterns.

Random Forest improved stability and predictive performance by averaging across multiple trees, reducing variance and improving generalisation. Its main limitation is reduced interpretability, as averaging across trees obscures individual feature interactions.

Both models relied on robust data preprocessing, including handling missing values, removal of unrealistic observations and One-to-Many encoding of categorical variables. Stratified cross-validation ensured reliable performance estimates, addressing class imbalance in the dataset.

6.3 FUTURE IMPROVEMENTS

Future work could focus on hyperparameter optimisation for the Random Forest, such as adjusting the number of trees, maximum tree depth and feature sampling at splits, to further enhance performance. Feature engineering may also improve model accuracy, for example by

combining service ratings into composite scores or deriving interaction terms (Salim & Lodin, 2023).

Alternative modelling approaches could be explored, including gradient boosting, support vector machines or neural networks, to assess potential predictive gains. Collecting additional demographic or behavioural data could further enhance the models' ability to capture the determinants of passenger satisfaction and improve actionable insights (Wang & Dumlao, 2025).

Overall, the study demonstrates the trade-off between interpretability and predictive performance. Decision Trees provide actionable insights, whereas ensemble methods maximise accuracy and robustness. These findings support targeted service improvements based on travel class and service quality.

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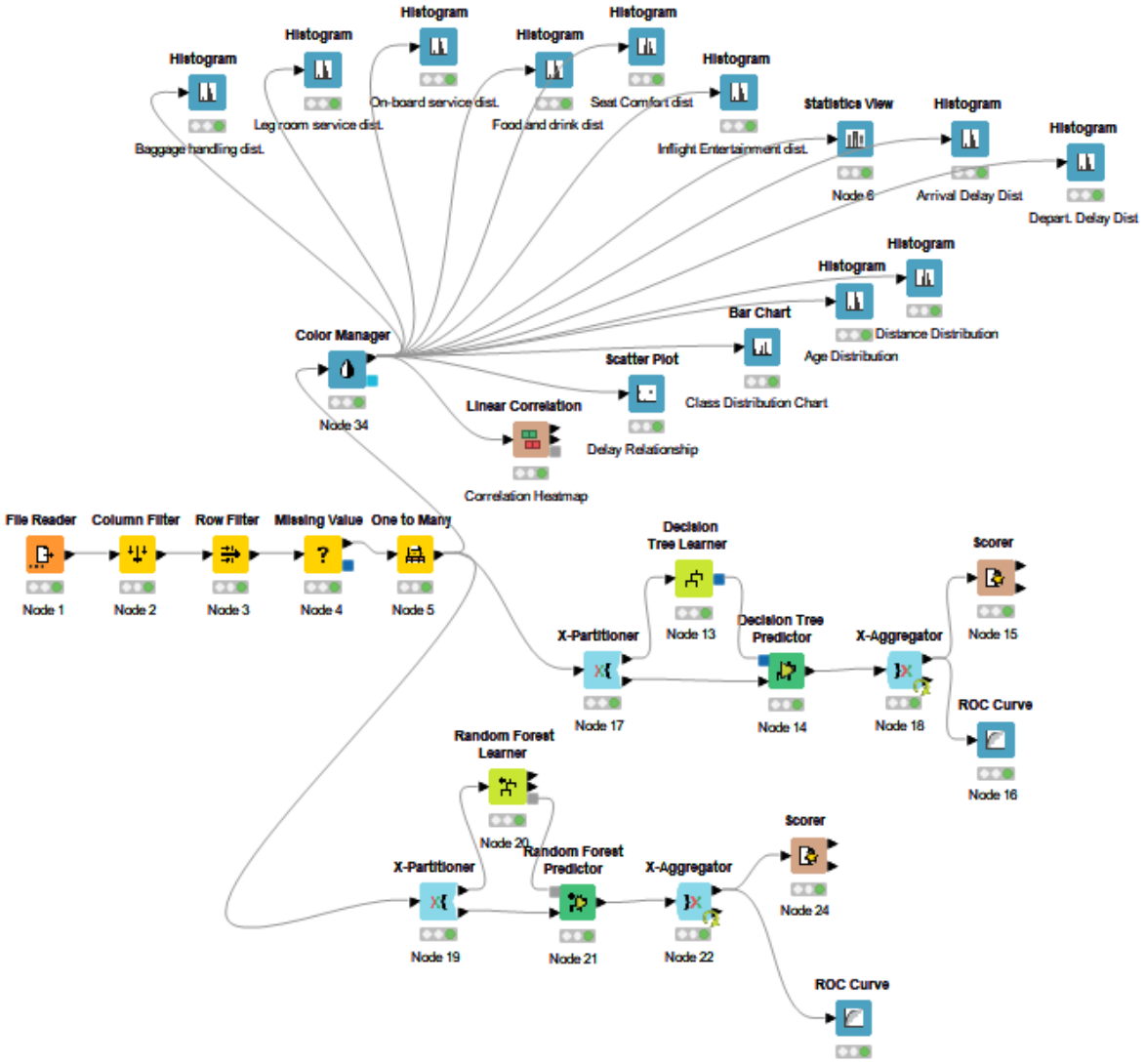
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8 APPENDIX

8.1 APPENDIX A: FULL WORKFLOW SCREENSHOT

Figure A1

Full KNIME workflow screenshot



Note. This figure shows the complete KNIME workflow used for data preprocessing, modelling, and evaluation.

8.2 APPENDIX B: GENERATIVE AI USE STATEMENT

Generative AI was used to support workflow verification and proofreading. Specifically, ChatGPT (OpenAI, GPT-5) assisted with checking the structure of the KNIME workflow and reviewing spelling, punctuation and grammar in the report.

8.2.1 Workflow Check

Prompt: "I have a knime workflow that includes one to many encoding, missing value imputation and stratified 5 fold cross-validation can you suggest if this setup is suitable for predicting airline passenger satisfaction"

Output: "Your workflow is appropriate for a classification task. One-to-Many encoding correctly converts categorical variables. Mean/mode imputation handles missing data adequately. Stratified 5-fold cross-validation ensures reliable performance evaluation despite class imbalance. Consider setting minimum records per node in your Decision Tree to prevent overfitting."

8.2.2 Sentence Proofreading

Prompt: "check this sentence for grammar and clarity: This approach maintained the dataset size while reducing the impact of missing data on model performance."

Output: "The sentence is grammatically correct and clear. For smoother readability, you could also phrase it as: 'This approach preserved the dataset size while minimising the impact of missing data on model performance.'"

8.2.3 Paragraph Proofreading

Prompt: "Check if this paragraph from my report is grammatically correct...."

Output: "The paragraph is clear. Minor improvements: ensure consistent tense, remove repeated words, and clarify technical terms. For example, change 'Many machine learning algorithms require numeric inputs, necessitating conversion of categorical attributes' to 'Most machine learning algorithms require numeric inputs, so categorical attributes were converted using One-to-Many encoding.'"